

Modeling: Introduction

DATA 202 21FA

Logistics

- Midterm 1 Online Portion is on Moodle
 - Don't forgot to copy-paste responses and completion code to Moodle
 - Closes Wed evening
- Project Milestone 2 due Friday
- No quiz or Discussion this week

Q&A!

What kinds of things can functions return?

Anything you can assign to a variable. (numbers, strings, data frames, plots, ...)

Are `as.numeric` and `as.integer` the same?

```
as.numeric("1.7")
```

```
[1] 1.7
```

```
as.integer("1.7")
```

```
[1] 1
```

```
as.integer("One Point Seven")
```

```
Warning: NAs introduced by coercion
```

```
[1] NA
```

2016 Honda Odyssey

EX-L 4dr Minivan (3.5L 6cyl 6A)



Mileage	37,183
Condition	Outstanding
Exterior	Modern Steel Metallic

Your appraisal As of 10/26/2019

	Trade-In	Private Party	Dealer Retail
Email report	\$21,645	\$23,731	
		\$25,666	
National Base Price ⓘ	\$19,676	\$21,694	\$23,484
Color Adjustment ⓘ	-\$43	-\$48	-\$52
Regional Adjustment ⓘ	\$103	\$114	\$123
Mileage Adjustment ⓘ	\$680	\$680	\$680
Condition Adjustment ⓘ	\$1,229	\$1,291	\$1,431
Value	\$21,645	\$23,731	\$25,666

What data frame does Edmunds have? Sketch an example. (What are the columns? What does each row represent?)

Missing Data

- Tools so far very good when we have all the data we want.
- Usually we *don't*-- or even we *can't*.

Different kinds of missingness:

1. How much will this car / house sell for?
2. What will Bitcoin trade for at end-of-day today?
3. What will this addition do to the price of my house?
4. I feel rotten. Is it Covid?
5. What word will the user type next on their keyboard?
6. Will this home-buyer default on their loan?

Discussion:

1. For each of these, discuss **what data is missing** (*why?*).
2. Which of these situations are similar to each other?

Some terminology

- **Supervised learning**: for each item independently, fill in an unobserved variable
 - **Regression**: fill in a *number*
 - **Classification**: fill in a *choice*
- **Unsupervised learning**: identify *relationships* between items
- **Forecasting**: predict how a sequence will continue (future observations)
- **Statistical inference**: fill in summary statistics (we wanted a population but only got a sample)
- **Causal inference**: fill in counterfactuals (what if?)

Predictive Analytics

Mostly *supervised learning* (*regression* and *classification*), some *forecasting*.

- A powerful tool to turn data into action.
- It works because God made the universe predictable (and successful prediction rewarding)
- **Need for wisdom:** It can be used for great good and great harm

Power of Predictive Modeling

- **Medicine:** wearable monitor for seizures or falls, detect malaria from blood smears, find effective drug regimens from medical records
- **Drug Discovery:** predict the efficacy of a synthesis plan for a drug
- **Precision Agriculture:** predict effect of micro-climate on plant growth
- **Urban Planning:** forecast resource needs, extreme weather risks, ...
- **Government:** classify feedback from constituents
- **Retail:** predict items in a grocery order
- **Recommendation systems:** Amazon, Netflix, YouTube, ...
- **User interfaces:** gesture typing, autocomplete / autocorrect

and so much more...

The universe is surprisingly predictable

- God created the world with actionable structure
 - We gradually learn how to perceive that structure and act within it.
 - The better our perceptions align with how the universe is structured, the better our actions
 - We can discover that structure by learning to be less surprised by what we see (= predicting our perceptions)
- Perceptions are thus both accurate and fallable.

Predictive modeling technology: Need for wisdom

- Potential for great good
- But also great harm:
 - Lack of **fairness** in facial recognition, sentencing, lending, job applicant scoring, ...
 - Lack of **transparency** in how "Big Data" systems make conclusions
 - Lack of **privacy** as data is increasingly collected and aggregated
 - Amplification of extreme positions in social media, YouTube, etc.
 - Oversimplification of human experience
 - Hidden human labor
 - Illusion of objectivity
 - ...!

Example: Home Sale Prices

From Ames, Iowa home sales, 2006-2010. (De Cock, 2011)

Lot_Area	Total_Bsmt_SF	Gr_Liv_Area	Garage_Cars	Sale_Price
31770	1080	1656	2	215000
11622	882	896	1	105000
14267	1329	1329	1	172000
11160	2110	2110	2	244000
13830	928	1629	2	189900

(2930 total rows)

- *Y: response variable (aka outcome, dependent variable):*
Sale_Price
- *X: features (aka predictors, covariates, etc.):* everything else

Note: X is much easier to measure than Y

Model

Predictions

